

UNIVERSITY OF BRISTOL

Faculty of Science

**MOCK EXAMINATION — SOLUTIONS AND
MARK SCHEME****Core Physics I — Classical, Quantum and Thermal Physics**

Module Code: PHYS10012 — Paper 2

Total Marks: 100

Marking Notes:

- Mark allocations are shown in **blue** at each step.
 - Final answers are highlighted in green boxes.
 - “*(From formula sheet)*” indicates the equation is on the provided formula sheet.
 - “*(Recalled — not on formula sheet)*” indicates the equation must be known from memory.
 - Award partial marks for correct method even if the final answer is wrong.
 - Accept equivalent correct forms of answers.
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Section A: Oscillations and Waves — Answers

1. C — From $x(t) = A \cos(\omega_0 t)$, we identify $A = 0.04$ m and $\omega_0 = 10\pi$ rad/s. The acceleration is $a(t) = -\omega_0^2 x(t)$, so the maximum acceleration is $|a_{\max}| = \omega_0^2 A = (10\pi)^2 \times 0.04 = 100\pi^2 \times 0.04 = 4\pi^2 \approx 39.5$ m/s². (*Recalled — not on formula sheet*)

2. C — Total energy $E = \frac{1}{2}kA^2$. At displacement x : PE = $\frac{1}{2}kx^2$ and KE = $E - \text{PE} = \frac{1}{2}k(A^2 - x^2)$. Setting KE = PE: $\frac{1}{2}k(A^2 - x^2) = \frac{1}{2}kx^2$, so $A^2 = 2x^2$, giving $x = A/\sqrt{2}$. (*Recalled — not on formula sheet*)

3. A — The energy in a damped oscillator decays as $E(t) = E_0 e^{-\gamma t}$. After n oscillations, $t = nT = 2\pi n/\omega_0$. Setting $E/E_0 = 1/2$: $e^{-\gamma(2\pi n/\omega_0)} = 1/2$, so $2\pi n\gamma/\omega_0 = \ln 2$. Since $Q = \omega_0/\gamma$ (*From formula sheet*), we get $n = Q \ln 2/(2\pi) = 50 \times 0.693/6.283 \approx 5.5$. The answer is **A**.

4. C — At resonance $\omega = \omega_0$, so $\tan \delta = \gamma\omega_0/(\omega_0^2 - \omega_0^2) = \gamma\omega_0/0 \rightarrow \infty$, which gives $\delta = \pi/2$. The displacement lags the driving force by $\pi/2$. (*From formula sheet*)

5. B — Option B: $y = Ae^{-(x-vt)^2}$ is a function of $(x - vt)$ only. Any function $f(x - vt)$ satisfies the wave equation, since $\partial^2 f/\partial x^2 = f''$ and $(1/v^2)\partial^2 f/\partial t^2 = f''$. Option A: $y = A \sin(kx) \cos(kx) = (A/2) \sin(2kx)$ has no time dependence, so $\partial^2 y/\partial t^2 = 0$ but $\partial^2 y/\partial x^2 \neq 0$; it does not satisfy the wave equation. Option C: $y = Ae^x \cos(vt)$ gives $\partial^2 y/\partial x^2 = Ae^x \cos(vt)$ and $(1/v^2)\partial^2 y/\partial t^2 = -Ae^x \cos(vt)$; these are not equal, so it fails. Option D: $y = A \sin(kx + \omega t^2)$ has $\partial^2 y/\partial t^2$ that produces terms proportional to $\cos(kx + \omega t^2)$ and $\sin(kx + \omega t^2)$ with time-dependent coefficients, so it does not satisfy the wave equation. (*From formula sheet*)

6. B — $r = (Z_1 - Z_2)/(Z_1 + Z_2) = (0.5 - 1.5)/(0.5 + 1.5) = -1.0/2.0 = -1/2$. The negative sign indicates phase inversion upon reflection (going from a lighter to a heavier string). (*From formula sheet*)

7. B — From $\beta = 10 \log_{10}(I/I_0)$: $I/I_0 = 10^{\beta/10} = 10^{85/10} = 10^{8.5} = \sqrt{10} \times 10^8 = 3.162 \times 10^8$. Hence $I = 3.162 \times 10^8 \times 10^{-12} = 3.16 \times 10^{-4}$ W/m². (*From formula sheet*)

8. A — Source receding: $f' = f \frac{v}{v + v_s} = 800 \times \frac{340}{340 + 40} = 800 \times \frac{340}{380} = 800 \times 0.8947 = 715.8 \approx 716$ Hz. (*From formula sheet*)

9. C — For an open-closed pipe, only odd harmonics are present: $f_n = nv/(4L)$ for $n = 1, 3, 5, 7, 9, \dots$ (*From formula sheet*)

10. C — The group velocity $v_g = d\omega/dk$ represents the speed at which the envelope (and energy) of the wave packet propagates. In a dispersive medium, the phase velocity $v_p = \omega/k$ depends on k , so generally $v_g \neq v_p$. The group velocity may be greater, equal, or less than the phase velocity depending on the dispersion relation. (*From formula sheet*)

Section B: Quantum Physics — Answers

1. C — $\langle\phi|\psi\rangle = \langle\phi|\psi\rangle$. Taking the conjugate of $|\phi\rangle$: $\langle\phi| = \frac{-i}{\sqrt{2}}\langle 1| + \frac{1}{\sqrt{2}}\langle 2|$. Then $\langle\phi|\psi\rangle = \frac{-i}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cdot \frac{i}{\sqrt{2}} = \frac{-i}{2} + \frac{i}{2} = 0$. The two states are orthogonal. (*Recalled — not on formula sheet*)

2. B — Check: $|\frac{1}{\sqrt{2}}|^2 + |\frac{i}{\sqrt{2}}|^2 = \frac{1}{2} + \frac{1}{2} = 1$. Option A gives $1/4 + 1/4 = 1/2 \neq 1$. Option C gives $1 + 1 = 2 \neq 1$. Option D gives $1/3 + 1/3 = 2/3 \neq 1$. (*Recalled — not on formula sheet*)

3. B — $\langle\alpha|\beta\rangle = \frac{1}{\sqrt{2}} \cdot c + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{c}{\sqrt{2}} + \frac{1}{2}$. For orthogonality, set this to zero: $c/\sqrt{2} = -1/2$, so $c = -1/\sqrt{2}$. (*Recalled — not on formula sheet*)

4. B — A unitary operator preserves inner products and norms. The defining property is $U^\dagger U = U U^\dagger = I$. Option A ($U^\dagger = U$) is the condition for a Hermitian operator. Option C ($U^2 = I$) would make U an involution, which is not required. Option D ($U = U^{-1}$) is the same as C. (*Recalled — not on formula sheet*)

5. B — $\langle S_z \rangle = \langle\psi| S_z |\psi\rangle$. Using $S_z = \frac{\hbar}{2}(|\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow|)$: $\langle S_z \rangle = \frac{\hbar}{2}(|\frac{1}{\sqrt{3}}|^2 - |\sqrt{2/3}|^2) = \frac{\hbar}{2}(\frac{1}{3} - \frac{2}{3}) = \frac{\hbar}{2} \times (-\frac{1}{3}) = -\frac{\hbar}{6}$. (*From formula sheet*)

6. C — $|\uparrow_x\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle + |\downarrow_z\rangle)$. The probability of measuring $S_z = +\hbar/2$ is $|\langle\uparrow_z|\uparrow_x\rangle|^2 = |1/\sqrt{2}|^2 = 1/2$. (*From formula sheet*)

7. C — For a stationary state (energy eigenstate), $|\psi(t)\rangle = e^{-iE_n t/\hbar} |E_n\rangle$. The overall phase factor cancels in all probability calculations: $P(a_k) = |\langle a_k | e^{-iE_n t/\hbar} |E_n\rangle|^2 = |\langle a_k | E_n \rangle|^2$, which is time-independent. Hence all measurement probabilities remain constant. (*From formula sheet*)

8. C — Option C can be factored: $\frac{1}{2}(|\uparrow\rangle_1 + |\downarrow\rangle_1)(|\uparrow\rangle_2 - |\downarrow\rangle_2)$. This is explicitly a product state $|a\rangle_1 |b\rangle_2$. Options A, B, and D cannot be written as product states (they are entangled). (*Recalled — not on formula sheet*)

9. B — Bosons have symmetric wave functions under particle exchange. The SWAP operator applied to a bosonic state gives $\text{SWAP} |\Psi\rangle = +|\Psi\rangle$. (Fermions give -1 .) (*Recalled — not on formula sheet*)

10. D — Beta decay ($n \rightarrow p + e^- + \bar{\nu}_e$) involves a down quark changing to an up quark via emission of a W^- boson. This is mediated by the weak nuclear force. (*Recalled — not on formula sheet*)

Section C: Long Answer Solutions

Question 1.

[15 marks]

Rotational Dynamics and Atwood Machine

(a) Rolling cylinder — energy conservation 3

The cylinder rolls without slipping, so $v = R\omega$. [1]

By conservation of energy, the gravitational PE lost equals the sum of translational and rotational KE:

$$Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}Mv^2 + \frac{1}{2}\left(\frac{1}{2}MR^2\right)\left(\frac{v}{R}\right)^2 = \frac{1}{2}Mv^2 + \frac{1}{4}Mv^2 = \frac{3}{4}Mv^2$$

[1]

Solving for v :

$$v = \sqrt{\frac{4gh}{3}} = \sqrt{\frac{4 \times 9.81 \times 3}{3}} = \sqrt{39.24} = 6.26 \text{ m/s}$$

[1]

$$v = \sqrt{4gh/3} = 6.26 \text{ m/s}$$

(b) Rolling cylinder — Newton's second law 3

Translational equation (along the slope): $Ma = Mg \sin 30^\circ - f$ [1]

where f is the friction force. Rotational equation about the centre of mass: $I\alpha = fR$, i.e. $\frac{1}{2}MR^2 \cdot \frac{a}{R} = fR$, giving $f = \frac{1}{2}Ma$. [1]

Substituting: $Ma = Mg \sin 30^\circ - \frac{1}{2}Ma$, so $\frac{3}{2}Ma = \frac{1}{2}Mg$, giving:

$$a = \frac{g}{3} = \frac{9.81}{3} = 3.27 \text{ m/s}^2$$

[1]

$$a = g/3 = 3.27 \text{ m/s}^2$$

(c) Atwood machine — massless pulley 4

For mass m_1 (heavier, moving down): $m_1g - T = m_1a$ [1]

For mass m_2 (lighter, moving up): $T - m_2g = m_2a$ [1]

Adding the two equations: $(m_1 - m_2)g = (m_1 + m_2)a$ (*From formula sheet*)

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2} = \frac{(6 - 4) \times 9.81}{6 + 4} = \frac{19.62}{10} = 1.962 \text{ m/s}^2$$

[1]

Substituting back: $T = m_1(g - a) = 6(9.81 - 1.962) = 6 \times 7.848 = 47.09 \text{ N}$.

Alternatively: $T = \frac{2m_1m_2g}{m_1 + m_2} = \frac{2 \times 6 \times 4 \times 9.81}{10} = 47.09 \text{ N}$. [1]

$$a = 1.962 \text{ m/s}^2, T = 47.09 \text{ N}$$

(d) **Atwood machine — pulley with inertia 5**

Let T_1 be the tension on the m_1 side and T_2 on the m_2 side.

For m_1 (moving down): $m_1g - T_1 = m_1a$ [1]

For m_2 (moving up): $T_2 - m_2g = m_2a$ [1]

For the pulley (solid disk, $I = \frac{1}{2}M_pR_p^2$), using $\tau = I\alpha$ and the no-slip condition $\alpha = a/R_p$: [1]

$$(T_1 - T_2)R_p = I\alpha = \frac{1}{2}M_pR_p^2 \cdot \frac{a}{R_p} \implies T_1 - T_2 = \frac{1}{2}M_pa$$

Adding all three equations: [1]

$$(m_1 - m_2)g = \left(m_1 + m_2 + \frac{M_p}{2}\right)a$$

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2 + M_p/2} = \frac{(6 - 4) \times 9.81}{6 + 4 + 1} = \frac{19.62}{11} = 1.784 \text{ m/s}^2$$

[1]

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2 + M_p/2} = 1.784 \text{ m/s}^2$$

Question 2.

[15 marks]

Ideal Gas, Carnot Engine, and Entropy**(a) Volume and internal energy 3**From the ideal gas law (*From formula sheet*): $PV = nRT$

$$V = \frac{nRT}{P} = \frac{3 \times 8.314 \times 400}{2 \times 10^5} = \frac{9976.8}{200000} = 0.04988 \text{ m}^3 \approx 0.0499 \text{ m}^3$$

[2]

Internal energy (diatomic gas, 5 degrees of freedom):

$$U = \frac{5}{2}nRT = \frac{5}{2} \times 3 \times 8.314 \times 400 = 24,942 \text{ J} \approx 24.9 \text{ kJ}$$

[1]

$$V = 0.0499 \text{ m}^3, U = 24.9 \text{ kJ}$$

(b) Adiabatic compression to half volume 4Using $TV^{\gamma-1} = \text{const}$ (*From formula sheet*), with $V_2 = V_1/2$ and $\gamma = 7/5$:

$$T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1} = 400 \times 2^{0.4}$$

[1]

Computing $2^{0.4}$: $2^{0.4} = e^{0.4 \ln 2} = e^{0.2773} = 1.3195$

$$T_2 = 400 \times 1.3195 = 527.8 \text{ K}$$

[1]

Using $PV^\gamma = \text{const}$ (*From formula sheet*):

$$P_2 = P_1 \left(\frac{V_1}{V_2} \right)^\gamma = 2 \times 10^5 \times 2^{1.4}$$

[1]

Computing $2^{1.4} = 2^1 \times 2^{0.4} = 2 \times 1.3195 = 2.639$:

$$P_2 = 2 \times 10^5 \times 2.639 = 5.278 \times 10^5 \text{ Pa}$$

[1]

$$T_2 = 527.8 \text{ K}, P_2 = 5.28 \times 10^5 \text{ Pa}$$

(c) Carnot engine 4Carnot efficiency (*From formula sheet*):

$$\eta = 1 - \frac{T_c}{T_h} = 1 - \frac{300}{600} = 0.5 = 50\%$$

[1]

Work done per cycle (*Recalled — not on formula sheet*):

$$W = \eta Q_h = 0.5 \times 2000 = 1000 \text{ J}$$

[1]

Heat rejected (*Recalled — not on formula sheet*)($Q_h = W + Q_c$):

$$Q_c = Q_h - W = 2000 - 1000 = 1000 \text{ J}$$

[1]

Check: $Q_c/Q_h = 1000/2000 = T_c/T_h = 300/600 = 0.5 \checkmark$

[1]

$$\eta = 50\%, W = 1000 \text{ J}, Q_c = 1000 \text{ J}$$

(d) **Entropy changes 4**

Entropy change of the copper block (*From formula sheet*):

$$\Delta S_{\text{Cu}} = mc \ln \left(\frac{T_f}{T_i} \right) = 2 \times 385 \times \ln \left(\frac{300}{500} \right) = 770 \ln(0.6)$$

$$\ln(0.6) = -0.5108, \quad \Delta S_{\text{Cu}} = 770 \times (-0.5108) = -393.3 \text{ J/K}$$

[1]

Heat released by the copper:

$$Q = mc(T_i - T_f) = 2 \times 385 \times (500 - 300) = 770 \times 200 = 154,000 \text{ J}$$

Entropy change of the reservoir (*Recalled — not on formula sheet*):

$$\Delta S_{\text{res}} = \frac{Q}{T_R} = \frac{154,000}{300} = 513.3 \text{ J/K}$$

[1]

Total entropy change of the universe:

$$\Delta S_{\text{total}} = \Delta S_{\text{Cu}} + \Delta S_{\text{res}} = -393.3 + 513.3 = 120.0 \text{ J/K}$$

[1]

Comment: $\Delta S_{\text{total}} > 0$, which is consistent with the second law of thermodynamics. The process is irreversible (there is a finite temperature difference between the copper and the reservoir), so the total entropy of the universe must increase.

[1]

$$\Delta S_{\text{Cu}} = -393.3 \text{ J/K}, \Delta S_{\text{res}} = +513.3 \text{ J/K}, \Delta S_{\text{total}} = +120.0 \text{ J/K} > 0$$

Question 3.

[15 marks]

Waves on Strings and Sound**(a) String properties 3**

Linear mass density:

$$\mu = \frac{m}{L} = \frac{0.012}{1.5} = 0.008 \text{ kg/m}$$

[1]

Wave speed (*From formula sheet*):

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{75}{0.008}} = \sqrt{9375} = 96.82 \text{ m/s}$$

[1]

Fundamental frequency (string fixed at both ends) (*From formula sheet*):

$$f_1 = \frac{v}{2L} = \frac{96.82}{2 \times 1.5} = \frac{96.82}{3.0} = 32.27 \text{ Hz}$$

[1]

$$\mu = 0.008 \text{ kg/m}, v = 96.82 \text{ m/s}, f_1 = 32.27 \text{ Hz}$$

(b) Harmonic wave at 3rd harmonic 3

The 3rd harmonic frequency:

$$f_3 = 3f_1 = 3 \times 32.27 = 96.82 \text{ Hz}$$

[1]

Angular frequency and wavenumber (*Recalled — not on formula sheet*):

$$\omega = 2\pi f_3 = 2\pi \times 96.82 = 608.3 \text{ rad/s}$$

The 3rd harmonic wavelength: $\lambda_3 = 2L/3 = 2 \times 1.5/3 = 1.0 \text{ m}$

$$k = \frac{2\pi}{\lambda_3} = \frac{2\pi}{1.0} = 6.283 \text{ rad/m}$$

[1]

The travelling wave equation (*From formula sheet*):

$$y(x, t) = A \sin(kx - \omega t) = 0.002 \sin(6.28x - 608.3t) \text{ m}$$

[1]

$$y(x, t) = 0.002 \sin(6.28x - 608.3t) \text{ m, with } \omega = 608.3 \text{ rad/s and } k = 6.28 \text{ rad/m}$$

(c) Impedance and reflection/transmission 4

Impedance of string 1 (*From formula sheet*):

$$Z_1 = \mu_1 v_1 = 0.008 \times 96.82 = 0.7746 \text{ kg/s}$$

[1]

Wave speed in string 2: $v_2 = \sqrt{T/\mu_2} = \sqrt{75/0.032} = \sqrt{2343.75} = 48.41 \text{ m/s}$

Impedance of string 2:

$$Z_2 = \mu_2 v_2 = 0.032 \times 48.41 = 1.549 \text{ kg/s}$$

[1]

Amplitude reflection coefficient (*From formula sheet*):

$$r = \frac{Z_1 - Z_2}{Z_1 + Z_2} = \frac{0.7746 - 1.549}{0.7746 + 1.549} = \frac{-0.7746}{2.324} = -\frac{1}{3}$$

[1]

Amplitude transmission coefficient (*From formula sheet*):

$$t = \frac{2Z_1}{Z_1 + Z_2} = \frac{2 \times 0.7746}{2.324} = \frac{1.549}{2.324} = \frac{2}{3}$$

[1]

Note: $Z_2 = 2Z_1$ exactly (since $\mu_2/\mu_1 = 4$ and $v_2/v_1 = 1/2$, so $Z_2/Z_1 = 2$), giving the clean fractions $r = -1/3$ and $t = 2/3$.

$$Z_1 = 0.775 \text{ kg/s}, Z_2 = 1.549 \text{ kg/s}, r = -1/3, t = 2/3$$

(d) Power reflection and transmission 2

Power reflection coefficient:

$$R = r^2 = \left(-\frac{1}{3}\right)^2 = \frac{1}{9} \approx 0.111$$

[1]

Power transmission coefficient: (*Recalled — not on formula sheet*)

$$\mathcal{T} = 1 - R = 1 - \frac{1}{9} = \frac{8}{9} \approx 0.889$$

Verification: $\mathcal{T} = \frac{Z_2}{Z_1} t^2 = 2 \times \left(\frac{2}{3}\right)^2 = 2 \times \frac{4}{9} = \frac{8}{9} \checkmark$

$R + \mathcal{T} = 1/9 + 8/9 = 1 \checkmark$

[1]

$$R = 1/9 \approx 0.111, \mathcal{T} = 8/9 \approx 0.889, R + \mathcal{T} = 1$$

(e) **Open-closed organ pipe 3**

For an open-closed pipe, only odd harmonics are present (*From formula sheet*):

$$f_n = \frac{nv_s}{4\ell}, \quad n = 1, 3, 5, \dots$$

[1]

First resonant mode ($n = 1$):

$$f_1 = \frac{340}{4 \times 0.85} = \frac{340}{3.4} = 100 \text{ Hz}$$

[1]

Second resonant mode ($n = 3$): $f_3 = 3 \times 100 = 300 \text{ Hz}$

Third resonant mode ($n = 5$): $f_5 = 5 \times 100 = 500 \text{ Hz}$

[1]

$$f_1 = 100 \text{ Hz}, f_3 = 300 \text{ Hz}, f_5 = 500 \text{ Hz}$$

Question 4.

[15 marks]

Mach-Zehnder Interferometer and Two-Particle States**(a) MZ interferometer — no phase shifter 4**

After the first beam splitter (BS1):

$$|T\rangle \rightarrow \frac{1}{\sqrt{2}}(|T\rangle + i|B\rangle)$$

[1]

After the mirrors (each path picks up a phase factor of i):

$$\frac{1}{\sqrt{2}}(i|T\rangle + i \cdot i|B\rangle) = \frac{1}{\sqrt{2}}(i|T\rangle - |B\rangle)$$

[1]

After the second beam splitter (BS2), using $|T\rangle \rightarrow \frac{1}{\sqrt{2}}(|T\rangle + i|B\rangle)$ and $|B\rangle \rightarrow \frac{1}{\sqrt{2}}(i|T\rangle + |B\rangle)$:

$$\begin{aligned} & \frac{1}{\sqrt{2}} \left[i \cdot \frac{1}{\sqrt{2}}(|T\rangle + i|B\rangle) + (-1) \cdot \frac{1}{\sqrt{2}}(i|T\rangle + |B\rangle) \right] \\ &= \frac{1}{2} [i|T\rangle + i^2|B\rangle - i|T\rangle - |B\rangle] \\ &= \frac{1}{2} [i|T\rangle - |B\rangle - i|T\rangle - |B\rangle] \\ &= \frac{1}{2} [-2|B\rangle] = -|B\rangle \end{aligned}$$

[1]

The final state is $-|B\rangle$ (overall phase is unphysical).

$$P(T) = 0, \quad P(B) = |-1|^2 = 1$$

All photons exit through port B .

[1]

Final state: $-|B\rangle$. $P(T) = 0, P(B) = 1$.

(b) MZ with phase shifter 3

With phase shifter $e^{i\varphi}$ on the top path, the state before BS2 becomes:

$$\frac{1}{\sqrt{2}}(ie^{i\varphi}|T\rangle - |B\rangle)$$

[1]

After BS2:

$$\begin{aligned} & \frac{1}{\sqrt{2}} \left[ie^{i\varphi} \cdot \frac{1}{\sqrt{2}}(|T\rangle + i|B\rangle) + (-1) \cdot \frac{1}{\sqrt{2}}(i|T\rangle + |B\rangle) \right] \\ &= \frac{1}{2} [ie^{i\varphi}|T\rangle - e^{i\varphi}|B\rangle - i|T\rangle - |B\rangle] \\ &= \frac{1}{2} [i(e^{i\varphi} - 1)|T\rangle - (e^{i\varphi} + 1)|B\rangle] \end{aligned}$$

[1]

Detection probabilities:

$$P(T) = \frac{|e^{i\varphi} - 1|^2}{4} = \frac{(e^{i\varphi} - 1)(e^{-i\varphi} - 1)}{4} = \frac{2 - 2\cos\varphi}{4} = \frac{1 - \cos\varphi}{2} = \sin^2\left(\frac{\varphi}{2}\right)$$

$$P(B) = \frac{|e^{i\varphi} + 1|^2}{4} = \frac{2 + 2\cos\varphi}{4} = \frac{1 + \cos\varphi}{2} = \cos^2\left(\frac{\varphi}{2}\right)$$

$$\text{Check: } P(T) + P(B) = \sin^2(\varphi/2) + \cos^2(\varphi/2) = 1 \quad \checkmark$$

[1]

$$P(T) = \sin^2(\varphi/2), \quad P(B) = \cos^2(\varphi/2)$$

(c) **Entanglement proof 4**

Suppose $|\Psi\rangle$ can be written as a product state:

$$|\Psi\rangle = (\alpha|\uparrow\rangle + \beta|\downarrow\rangle)_1 \otimes (\gamma|\uparrow\rangle + \delta|\downarrow\rangle)_2$$

Expanding:

$$= \alpha\gamma|\uparrow\uparrow\rangle + \alpha\delta|\uparrow\downarrow\rangle + \beta\gamma|\downarrow\uparrow\rangle + \beta\delta|\downarrow\downarrow\rangle$$

[1]

Comparing with $|\Psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$, we need:

$$\alpha\gamma = 0, \quad \alpha\delta = \frac{1}{\sqrt{2}}, \quad \beta\gamma = -\frac{1}{\sqrt{2}}, \quad \beta\delta = 0$$

[1]

From $\alpha\gamma = 0$: either $\alpha = 0$ or $\gamma = 0$.

- If $\alpha = 0$: then $\alpha\delta = 0 \neq 1/\sqrt{2}$. Contradiction.
- If $\gamma = 0$: then $\beta\gamma = 0 \neq -1/\sqrt{2}$. Contradiction.

Therefore no product decomposition exists, and the state is entangled.

[1]

If S_z is measured on particle 1 and gives $+\hbar/2$ (i.e. outcome $|\uparrow\rangle_1$), the state collapses to the component of $|\Psi\rangle$ associated with $|\uparrow\rangle_1$, which is $|\downarrow\rangle_2$. Particle 2 is in state $|\downarrow\rangle_2$ ($S_z = -\hbar/2$).

[1]

The state is entangled (proof by contradiction above). After measuring $S_z = +\hbar/2$ on particle 1, particle 2 is in state $|\downarrow\rangle_2$.

(d) **Time evolution of spin state 4**

(i) The energy eigenstates and eigenvalues are $|\uparrow_z\rangle$ with $E_+ = E_0/2$, and $|\downarrow_z\rangle$ with $E_- = -E_0/2$. The initial state is:

$$|\psi(0)\rangle = |\uparrow_x\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle + |\downarrow_z\rangle)$$

Using the time evolution formula (*From formula sheet*):

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} \left(e^{-iE_0t/(2\hbar)} |\uparrow_z\rangle + e^{iE_0t/(2\hbar)} |\downarrow_z\rangle \right)$$

[1]

(ii) The probability of measuring $S_z = +\hbar/2$:

$$P(\uparrow_z) = \left| \frac{1}{\sqrt{2}} e^{-iE_0t/(2\hbar)} \right|^2 = \frac{1}{2}$$

The probability is constant: $P(\uparrow_z) = 1/2$ for all t .

[1]

This makes sense because $|\uparrow_z\rangle$ and $|\downarrow_z\rangle$ are energy eigenstates with equal weights $|c_n|^2 = 1/2$, which do not change in time.

(iii) To find $\langle S_x \rangle(t)$: (*From formula sheet*) $S_x = \frac{\hbar}{2}(|\uparrow\rangle\langle\downarrow| + |\downarrow\rangle\langle\uparrow|)$.

$$S_x |\psi(t)\rangle = \frac{\hbar}{2} \cdot \frac{1}{\sqrt{2}} \left(e^{-iE_0t/(2\hbar)} |\downarrow_z\rangle + e^{iE_0t/(2\hbar)} |\uparrow_z\rangle \right)$$

$$\begin{aligned} \langle S_x \rangle(t) &= \langle \psi(t) | S_x | \psi(t) \rangle \\ &= \frac{\hbar}{2} \cdot \frac{1}{2} \left(e^{iE_0t/(2\hbar)} e^{iE_0t/(2\hbar)} + e^{-iE_0t/(2\hbar)} e^{-iE_0t/(2\hbar)} \right) \\ &= \frac{\hbar}{4} \left(e^{iE_0t/\hbar} + e^{-iE_0t/\hbar} \right) = \frac{\hbar}{2} \cos\left(\frac{E_0t}{\hbar}\right) \end{aligned}$$

[2]

Check: at $t = 0$, $\langle S_x \rangle = \hbar/2$, which is correct since the state is $|\uparrow_x\rangle$. ✓

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} \left(e^{-iE_0t/(2\hbar)} |\uparrow_z\rangle + e^{iE_0t/(2\hbar)} |\downarrow_z\rangle \right)$$

$$P(\uparrow_z) = 1/2 \text{ (constant)}$$

$$\langle S_x \rangle(t) = \frac{\hbar}{2} \cos(E_0t/\hbar)$$